IntentTrace: Anchoring LLM-Generated Feedback in Design Intent for Design Judgment and Iteration of Works

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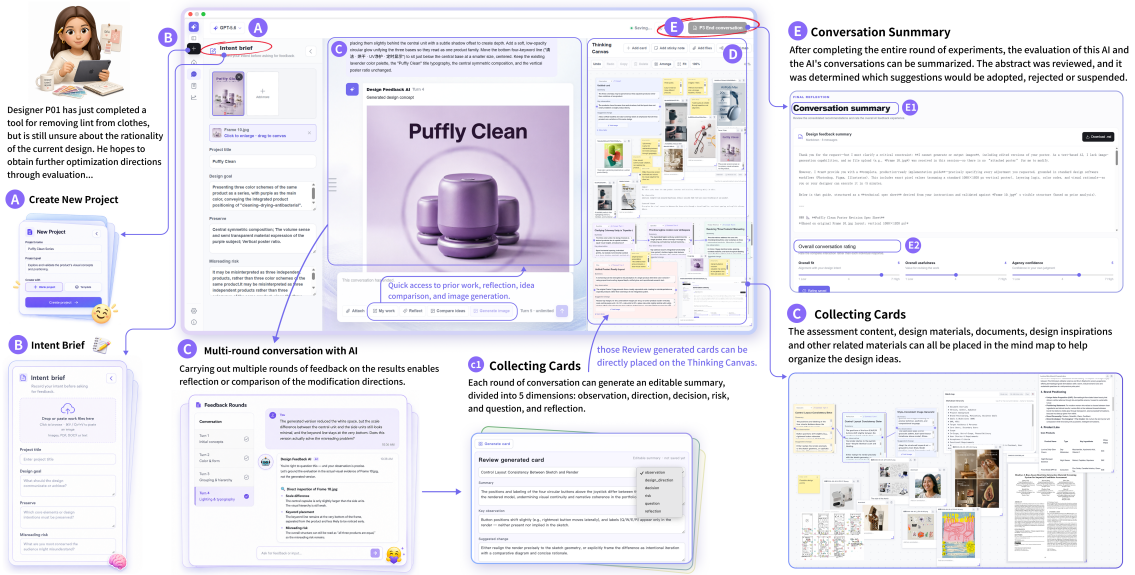


Fig. 1. The IntentTrace interface consists of six processes. (A) The designer creates the project and submits the design results; (B) Completes the intention description; (C) Conducts multiple rounds of feedback with the AI to facilitate reflection or comparison of modification directions; (C1) Each time the AI generates an evaluation, it can generate a summary card and support editing; (D) On the right side, there is a "mind map" that summarizes various design materials; (E) Clicking the end of the conversation; (E1) the AI will detail the summary of the conversation content and indicate which suggestions were adopted, rejected, or postponed; (E2) finally, it rates from three dimensions: "fit", "usefulness", and "confidence" (F) This interaction process is saved by the system backend. Specific details can be found in Section 4.

Evaluating artifacts in open-ended design tasks is challenging because criteria are often ambiguous and evolve through self-judgment, external feedback, and iterative revision. As large language models (LLMs) become part of design practice, they offer new opportunities to support artifact evaluation and refinement. This paper presents IntentTrace, an intent-tracing AI feedback system that helps designers align LLM-generated feedback with their original design intent across multiple rounds of refinement. IntentTrace integrates design intent, iterative AI feedback, evaluation-dimension anchoring, and revision reflection into a unified workflow. Through a controlled experiment and interviews, we examined designers' refinement challenges, AI use contexts, and responses to two feedback conditions: Anchor and No-Anchor. Results show that designers often turn to LLMs when revision directions are unclear, feedback needs concretization, or their own ideas require validation. However, the main challenge is not the lack of feedback, but the lack of stable grounds for judgment. Compared with the No-Anchor condition, the Anchor condition led to higher AI-independent self-efficacy and stronger perceived fit between AI feedback and design goals. Interviews further suggest that intent references help maintain continuity across feedback rounds. We argue that AI feedback systems should move beyond automatic evaluation toward creative design support that enables intent tracing, problem reflection, and judgment calibration.

CCS Concepts: • **Human-centered computing** → **HCI design and evaluation methods**; *Interactive systems and tools*; • **Computing methodologies** → Natural language processing.

Additional Key Words and Phrases: human-computer interaction, interface design, large language models, AI feedback, design judgment, design artifact evaluation

1 Introduction

Open-ended design is not a linear progression from a problem to a definitive solution; rather, design problems and solutions co-evolve [30], while designers continually revise their judgments through action and reflection [103]. Because open-ended problems have neither a single answer nor a clear stopping rule [99], the same artifact may be evaluated simultaneously in terms of visual expression, functional logic, context of use, and technical feasibility [26, 44]. Designers must judge not only whether a work is "good enough," but also which issue deserves priority at a given stage, which elements must be preserved, and whether a local revision would undermine the overall direction.

Design judgment also concerns design agency. Research across human–AI interaction and co-creation has supported user agency through adjustable levels of automation [93], iterative prompt refinement [88], and greater user influence over final artifacts [31]. Explanation and debugging interfaces can further make system behavior inspectable [109], expose mismatches between users' concepts and model interpretations, and support iterative correction of system behavior [46]. Taken together, this literature primarily treats agency as control over how AI intervenes, how inputs are specified, or how outputs are revised. Less is known about whether designers retain authority over evaluation criteria, problem definitions, and final trade-offs when AI begins to interpret problems and critique design work. We therefore ask:

RQ1: What judgment challenges do designers encounter when evaluating and revising their work and in what situations do they use LLMs?

External feedback can introduce new perspectives, but it does not automatically resolve difficulties in judgment. Structured and aggregated feedback can reveal recurring comments and conflicting criteria [75]; process records and iterative tools can support creative exploration [43, 105]; and comparison across perspectives can expose disagreements and boundary cases [94]. However, when suggestions are based on different standards, designers must still determine what problem each suggestion addresses, what criterion it invokes, whether it is consistent with their intention, and whether to adopt, partially adopt, reject, or defer it [56, 104]. Existing feedback systems can locate, aggregate and compare comments, but rarely preserve original intent, artifact evidence, evaluative rationales, and revision trade-offs within one interaction process[77, 79]. We therefore ask:

RQ2: How can designers continue to understand problems and form judgments from their own design intent while engaging with LLM feedback?

LLMs have recently expanded from content generation into artifact evaluation. Guideline-based LLMs can generate UI revision suggestions, although feedback utility may decline across iterations [36]; design-critique datasets can connect comments to specific visual regions [34]; vision–language models can compare UI quality and generate design guidance [122]; and visual grounding with iterative correction may improve where critiques are directed [32]. These studies establish the feasibility of automated design feedback, but primarily examine whether AI can generate critiques, whether model evaluations approximate expert judgments, and how feedback can be localized and presented.

AI evaluation remains constrained by how an artifact is represented, which criteria are predefined, and what domain context is available. Adding domain information does not ensure that users can identify LLM errors [55]. The explanation format shapes how recommendations are accepted [16], and trust changes with performance over repeated interactions [54]. Subjective trust and explanation preferences do not reliably predict actual performance [12], while visual explanations may foster inappropriate trust [130]. The Open-ended design lacks directly verifiable

correct answers. Even when designers reject specific suggestions proposed by AI, they may still accept its framing of the problem scope, priorities, or evaluation criteria. Prior work has seldom traced how designers decide whether feedback accords with their original intent or how multi-turn feedback changes their understanding of the problem, revision direction, and final authority. We therefore ask:

RQ3: Across multiple rounds of LLM feedback, how do designers interpret the AI's influence on their problem understanding, revision direction, and design autonomy?

To address these gaps, we first conducted formative interviews to identify the judgment challenges and situations in which designers use LLMs. We translated the findings into five design strategies – intention trajectories, a multimodal whiteboard, feedback decomposition, alignment verification, and reflection of judgment—and developed IntentTrace. In the Anchor condition, the LLM could access the participant's "Intent Brief"; in the No-Anchor condition, this information was withheld. Both conditions used identical evaluation dimensions, artifact materials, dialogue histories, feedback structures, and model parameters. Because prior work has not isolated whether an LLM's access to design intent changes the feedback experience, we ask:

RQ4: Compared with the No-Anchor condition, how does the Anchor condition affect perceived fit, willingness to adopt feedback, and confidence in one's judgment?

The formative interviews addressed RQ1 and informed five design strategies that were instantiated in IntentTrace (RQ2). After a design walkthrough, we conducted a single-factor, between-subjects experiment with 56 participants assigned equally to the Anchor and No-Anchor conditions. The sole independent variable was whether the LLM could access the Intent Brief; evaluation dimensions, artifact materials, dialogue history, feedback structure, and model parameters were held constant. System records, post-study questionnaires, and 12 interviews informed RQ2–RQ4. After correction for multiple comparisons, the only robust between-condition difference was in AI-independent self-efficacy.

This paper makes three contributions

(1) Through formative research, we reveal the specific difficulties designers encounter in maintaining judgment references, interpreting feedback fit, and tracing problem changes during LLM-assisted artifact evaluation. In doing so, we shift the focus from whether AI can provide suggestions to how designers form judgments about those suggestions.

(2) We propose and implement IntentTrace, which treats design intent as a visible, revisable, and verifiable interaction reference, while distinguishing AI observations, evaluative grounds, revision suggestions, and their associated trade-offs. This provides reusable design mechanisms for supporting reflective feedback.

(3) Through a single-factor condition comparison and in-depth interviews, we show how LLM access to design intent participates in problem clarification, frame shifts, revision trade-offs, and negotiations of design autonomy. We further distinguish surface-level fit, substantive understanding, and final adoption, providing empirical evidence for research on intent alignment and designer control in design-support AI systems.

2 Related Work

2.1 Designers' Judgment of Design Work

Design judgment is designers' capacity to make situated trade-offs among problems, solutions, and actions under incomplete information, conflicting goals, and the absence of a single correct answer [11, 26, 30, 103]. Nelson et al. regard judgment as the core of design action: rules and analysis can serve as resources; rules and analysis can provide resources, but they cannot replace the designer's decision about what is worth doing in a specific context [83].

Design goals, problem frames, alternatives, and evaluation criteria together provide reference points through which designers interpret design choices [4, 95]. Design rationale encompasses not only final decisions but also alternatives considered, criteria for trade-offs, and the argumentation through which decisions were reached [97, 116]. IBIS records design deliberation through an issue–position–argument structure [25, 63], while PHI uses hierarchical issues to decompose and trace complex problems [78, 106]. Later work has sought to document questions during the design process [9, 50]. or connect artifact features to grounds for judgment through feature–norm–evidence structures [117].

Human–AI co-creation research commonly operationalizes agency as control and autonomy, while also considering creative self-efficacy, ownership of outcomes, and the ability to regulate AI involvement [19, 49, 96, 140]. At the interaction level, such control includes choosing when AI intervenes [52], adjusting its initiative [63], accepting or rejecting suggestions[135], editing intermediate outputs, comparing alternatives[124], and selectively adopting results so that AI output moves closer to the designer's intent [96]. Agency in human–AI co-creation is not allocated once and for all; it shifts with the design stage [66], the AI's level of initiative [96], and the control mechanisms provided by the system [140].

2.2 Evaluation and Feedback in Open-Ended Design Practice

Open-ended design rarely has fixed answers. Evaluation criteria change with design goals, target audiences, and iteration stages, requiring continual trade-offs among goals, constraints, aesthetics, and contexts of use [83, 103]. Design judgment is therefore not a score assigned after completion, but an ongoing activity spanning problem framing, idea generation, comparison, and revision [30, 83, 103]. Problems and solutions co-evolve [3], while design briefs are continually reframed through negotiation, questioning, and experimentation [14, 95, 110].

Design feedback can turn comments into interpretable grounds for revision through explicit evaluation goals, open discussion, and post-feedback reflection [1, 9, 79, 97, 106]. Prior systems increase specificity through the selection of design principles, localized annotations, and criterion-based evaluation [76, 125, 128], or separate visual elements, attention order, first impressions, design conventions, and intended goals into distinct evaluation tasks [76, 125]. Feedback also arrives through peers, social net artifacts, and other channels [119, 131]. Corresponding interfaces support reviewing, comparing, and interpreting comments [132], posing open questions before displaying evaluative opinions [67], clustering feedback by design region and comparing it across sources [133], and turning feedback reception into operations such as comparison, simulated revision, and goal reframing [42, 58, 60, 137]. Together, these mechanisms suggest that feedback in iterative work should remain inspectable, rejectable, and revisable [38, 48].

Existing AI design-feedback systems organize issues and revisions through heuristic inspection [37], localized screenshot annotations [33, 35], bookmarks, highlights, or heatmaps [85, 138]. These systems have expanded from generating solutions to providing visual references, structured collaboration, and feedback across multiple stages of design [40, 121], thereby supporting the design process through visual referencing, structured collaboration, and multi-stage intervention [18, 20, 98]. This literature primarily examines output quality, creative performance, and design outcomes [18, 20, 98], as well as mechanisms that help designers understand AI [57, 118, 129], but less often traces how AI changes designers' definitions of the "main problem." We therefore focus on comparison against intent, rationales for feedback adoption, and authority over problem definition and final revisions.

2.3 LLMs as Critics and Evaluators of Design Work

As multimodal foundation models increasingly support joint processing of text and images and cross-modal reasoning [53, 134], generative AI has begun to move from solution generation into artifact evaluation [121, 142]. Existing

systems convert interface screenshots and descriptions [37, 123], layer hierarchies and spatial relationships [35, 37], or perceptual metrics [56, 104], into model-readable representations, and then generate ratings and revision suggestions using heuristic criteria [33], expert critiques [35], or design principles [69, 82].

Such evaluation is becoming embedded in ongoing design activity. Work at the concept stage has traced revisions made after designers receive ChatGPT suggestions [126] and compared design concepts and creative processes with and without ChatGPT [59]. Visualization tools combine design guidelines and perceptual metrics to generate natural-language suggestions [104] and support reevaluation after revision [56, 92], while sidebars, canvas annotations, and automated modifications present actionable feedback [68, 71]. Related mechanisms have been applied to architectural competition review [91], reflection and critique in design studios [100], and prediction for engineering prototypes [136]. AI agents can operate mobile prototypes [127] or live websites [73], and can combine personas, interface code, and interaction traces to identify usability issues [45]. Multi-agent systems can simulate website A/B tests [72] or user behavior and preferences [5], while multimodal models evaluate candidate interfaces through pairwise choice or absolute scores [74]. Although models are proficient at recognizing text and images, reliance on AI can increase inappropriate adoption as artifacts improve across iterations [13, 102].

Prior work emphasizes that designers must remain involved and correct AI misunderstandings of design goals [27], and that tasks requiring contextual interpretation and judgment should remain primarily under designers' control [139]. Yet most studies ask whether AI can generate design critiques [35, 37], whether its judgments approximate those of experts or users [56, 74], and how critiques should be presented [33, 68, 71]. They less often examine how designers decide whether a critique aligns with their original intent. Most systems support repeated evaluation during design without separately exposing what the model observed, which principle informed its evaluation, and what it recommends changing.

3 Formative Research

The formative study is used to answer RQ1 and to build an empirical basis for RQ2. At this stage, we will explore how designers judge whether the work needs to be modified; what stages are the most vulnerable to loss of direction; how they understand feedback from different sources; what circumstances LLMs are currently used in; and what should be provided and not replaced by feedback from AI.

3.1 Interview participants and processes

Using purpose-level recruitment, we interviewed 20 design learners, practitioners, product managers and design educators (P1-P20; 10 women, 10 men), covering areas such as interaction, visualization, industry, products, services, digital media, space and design research, and different frequencies of AI usage. Inclusion criteria: (a) All participants are 18 years of age or older. (b) Nearly one month of participation in an open design task. (c) Has used a text-based LLM or generated AI to participate in the creation of the work within a week. IntentTrace developers directly involved are not included. Participation is entirely voluntary and free of charge, and materials are converted and de-marked. The study is subject to ethical review (approval number: 202606RB031) and the sample information is provided in Appendix A.

The interviews were conducted in a semi-structured format, centered on judgment nodes in the design process, conflicting evaluation criteria and feedback, AI use context and recommendations, and discussed how the system should support material organization and final decision-making. We use thematic analysis to lay the basis for coding, feedback conflicts, AI interventions, and trust boundaries before cross-case comparisons are made, leading to five challenges and corresponding design needs. The interview outline is contained in Appendix A.

3.2 Five design challenges

3.2.1 C1: Unstable reference to judgment. Participants usually have no shortage of candidates; the challenge is deciding which direction is worth continuing. The weights of vision, demand, innovation, realizability, cost, and time change over time. Some depend on experience and intuition, but it is difficult to justify them; others are increasingly looking at them, and the range of options continues to expand. P14 summarizes the judgment as a balance between demand, realizability, and appearance, indicating that it is not merely a matter of aesthetic preference.

Existing feedback does not automatically stabilize this reference. New opinions may add evidence and may introduce new evaluation criteria. If the designer fails to record the target, preserve the boundary, and current priorities, it is difficult to judge whether a change has been made in a deeper direction or has been inadvertently replaced.

3.2.2 C2: Multiple-modular design materials and AI outputs are difficult to integrate. The workflow crosses inspirational platforms, documents, drawings, modeling, rendering, and layout tools, and is then submitted to AI. The P3 will view the website for a long time and save the information in the folder; the P10 and P13 projects also undergo continuous tool switching. The relationship between materials, versions, and AI outputs is largely dependent on individual memory.

This challenge is not a direct requirement of the phrase "needs a whiteboard" in an interview, but rather a combination of work streams: when work, intent, and reference material cannot be presented together, the designer re-selects and interprets the context every time he asks a question, and AI can only rely on the partially extracted material feedback. The cost of tool switching is not only operational time, but also a judgment of the continuing loss of context and reconstruction.

3.2.3 C3: Feedback is difficult to translate into comparable judgment. Views such as "a little more" are too abstract; specific opinions may also be operational without justification. The designer still must judge what the feedback looks at, what standards it uses, what it recommends changing, and what it costs. Otherwise, more feedback will only result in a longer list of changes.

Participants expect AI to help concretize and compare rather than draw conclusions. P6 refers to AI's "not necessarily accurate" information, which still needs to be checked with other platforms; the use of both positive and reverse questions on the same issue can be argued to the contrary. P19 expects AI to set out not only the directions but also the basis for comparison and recommendation. The interview materials indicate that the feedback needs to be searchable, verifiable and comparable, rather than automatically consolidated by the system into one optimal answer.

3.2.4 C4: Perception adhesion may mask LLM evaluation sources. Participants welcomed AI feedback on their work, but also questioned its fit, evidence base, and reliance on general design principles. P7 and P17 distinguished AI's usefulness for documentation, inspiration, and image generation from its credibility as an evaluator, while P4 questioned feedback that felt overactive and insufficiently critical. References to design elements or repeated user goals could create a sense that "it understands me," but designers still lacked clarity about whether the judgment came from visual evidence, their own stated intent, general rules, or linguistic alignment.

Perceived alignment is therefore not sufficient evidence of feedback quality. Systems that optimize only for individualized expression may increase acceptance while weakening scrutiny of the feedback basis. Designers should be able to verify whether feedback continues their objective, respects stated boundaries, or introduces new evaluative standards.

3.2.5 C5: Undetectable to judge changes and problems. Multiple rounds of feedback change the presentation of the issue, but this change is rarely recorded separately. The dialogue may start with "if there is value in the design solution"

and end with “how to adjust the hierarchy”; designers may not be able to indicate when the problem is replaced, why confidence changes, or who introduces priorities. Chat records keep messages, but do not give rise to problems and judgment tracks.

This invisibility relates directly to design autonomy. Designers do not need to exclude AI impacts, but rather to identify where they occur and make a clear decision on whether to accept the new problem framework. P14 It is hoped that AI will give reasons, cases, or data to judge, while retaining its own final judgment; other participants also describe the desired role with “second perspective” “checking for tasks” rather than “deciding for me”.

3.3 Designer uses LLM for five scenarios

The interview answered RQ1, and the designer assigned different roles to AI under different uncertainties

Understanding tasks and organizing design materials. Participants use LLM decomposition requirements to summarize research, organize keywords, or indicate information for the next steps. At this point, AI primarily lowers the cost of information organization, and the output still needs to be checked by external sources.

Sensitization and generation of multinodular materials. Designers require AI to generate conceptual orientation, images, stories, emotions, or visual references. P18 sums it up as research compilation, inspiration, and imagery; this phase is more tolerant of uncertainty, but designers are concerned about homogenization.

Finds entry points when the change in direction is uncertain. When the work has developed and goes on from nowhere, an LLM is used to point out possible problems or to establish priorities for change. P15 often pursues an external perspective when one or two rounds of self-revision are still at a standstill.

Concretize abstract feedback and ICP. The designer requested AI to convert evaluations such as “better” and “more uniform” into implementable operations or to compare the benefits and costs of the two modification paths.

Check your own ideas and check before delivery. Participants use AI to check text, tasks, logic, conditions of achievement, or existing judgments, but retain them for data, aesthetics, and practicality.

3.4 From Challenge to Five Design Needs

We have translated five challenges into design needs, as shown in Table 1:

Table 1. Design challenges, design needs and design objectives resulting from formative research

Forming a sexual challenge	Corresponding design requirement	Design objective
C1 Discretionary reference	DR1 Intention and judgement references traceable	Recording of objectives, retention of boundaries and risk of misunderstanding and allowing for subsequent changes
C2 Multimodular Material Dispersion	DR2 materials can be assembled, organized and shared	In the same space, with reference, version and AI output, the context is controlled by the designer
C3 Feedback is difficult to compare	DR3 Feedback is disaggregated, searchable and comparable	Differentiate work observations, evaluation bases, recommendations and trade-offs, supporting multiple directions
C4 Collapse basis opacity	DR4 Intent alignment and feedback based on validation	Show the target and boundary to which the recommendation corresponds, distinguishing between repetition, specificity and substance
C5 Discovery of change in judgement	DR5 Discretionary and decision-making power	Recording of adoption, rejection, suspension and change of issues to enable designers to confirm final direction

DR1: Make design intentions and judgment references traceable. AI co-creation systems need to support designers in continuously articulating, clarifying, and revising their design intent, while preserving a process record of how that intent evolves alongside the development of design solutions [66, 86, 112]. The AI feedback system needs to help designers externalize their current design objectives, what they want to keep, what they fear being misunderstood,

and what they use when making trade-offs and how they change their intentions, thus helping designers to judge whether the modification is a continuation of the original direction or whether the design direction has been proactively reoriented based on new evidence.

DR2: Enable multimodal materials to be assembled, organized, and integrated into the common context. Recent studies on LLM-based creativity support systems show that spatial canvases and structured templates can integrate user-provided materials and AI-generated outputs into a shared workspace, supporting grouping, hierarchical organization, contextual association, and cross-source comparison [111, 129]. In response to the fragmentation of various design materials (reference diagrams, research text, versions of artifacts, and AI output), the system needs to provide a unified material space that allows designers to combine, group, and compare different materials and establish a relationship between materials and current issues.

DR3: Enable disaggregated, searchable, and comparable feedback. In response to the abstraction of feedback, conflict, or lack of justification, the system needs to translate a full recommendation into a judgment that can be examined separately [62, 70], including what AI has observed, what dimensions have been evaluated, what actions are recommended, and what trade-offs this action may bring. The designer should also be able to continue to ask for reasons, compare multiple directions, and defer decisions when insufficient evidence is available. The objective is to support designers in forming their own trade-offs, rather than allowing the system to automatically assign multiple opinions to the only answer.

DR4: Make the alignment of intent and the basis for feedback verifiable. In response to LLM restatements, affirmations, and language-specific issues that may create a sense of homogeneity, the system needs to articulate the relationship between the recommendations and the design objectives, the retention of boundaries, and evidence of the work. The design objective is to enable designers to examine the basis on which this "understanding" is based and to correct AI's misinterpretation of intent [61, 64].

DR5: Allows for reflection on changes in judgment and retains the final decision of the designer. The system needs to document the understanding of the designer's issues around the key rounds, as well as the reasons for the adoption, partial adoption, rejection, or suspension of the recommendation, in response to the issue framework, changes in priorities, and a shift in judgment confidence in multiple rounds of feedback. Rethinking mechanisms should be neutral in suggesting changes, so that designers can identify where AI is affecting judgment and whether to accept the new problem framework [90, 129].

This will be followed by a presentation on how IntentTrace translates these needs into research probes and provides a detailed description of the system's functionality.

4 IntentTrace: Intended tracking AI feedback study probe

4.1 System positioning and corporate processes

IntentTrace was achieved as a whole-house web application. The frontend is built with React and TypeScript [7]and uses Vinext [24]to provide page and service routers compatible with the Next.js App Router [84]. Backend interfaces such as text feedback, user authentication, and research data preservation are achieved by TypeScript Route Handlers [41]. Through the OpenAI Chat API [3, 87], the system calls large language models that support text and image input, organizes design intentions, artifact materials, current issues, and historical dialogue as the context of the model, and records user input, AI feedback, matching validation, and revision reflection. The current prototype is used to concretize

five design strategies and to observe how designers express intent, organize material, follow up on recommendations, and explain changes.

4.2 Anchor mechanism based on evaluation dimensions

Open design artifacts differ in terms of objectives, circumstances of use, and maturity, making it difficult to use a single cross-cutting scale. To develop an LLM feedback framework suitable for this study, we conducted a literature review on open-ended design judgment, design evaluation, problem setting, user tasks, originality, visual expression, and implementation feasibility, and then synthesized the relevant evaluation criteria. This resulted in five evaluation dimensions (Table 2). Based on this, the text LLM for the IntentTrace system is presented in the appendix: B.1-B.3.

Table 2. Evaluation Dimensions for LLM-Based Design Artifact Assessment in IntentTrace

LLM evaluation dimension	Main findings	Theory sources
1.Intent and problem-setting	Whether the work responds to the original design objectives; AI changes the original problem framework; and whether there is a reasonable basis for such a reorganization	Dorst & Cross [30];Uluoğlu [113]
2. User assignments and relevance	Design fit for target users, tasks and specific use situations	Hornbæk [51]
3. Original and conceptual values	Whether the concept is innovative, relevant and of design value, rather than simply pursuing the formal "new"	O'Quin & Besemer [89];Sarkar & Chakrabarti [101];Christensen & Ball [23]
4. Visual expression and information communication	Whether visual organization, hierarchy and expression can effectively convey design concepts	Moshagen & Thielsch [81];Christensen & Ball [23]
5. Overall coherence, completion and enforceability	Whether the components form a complete programme; whether design decisions are coordinated; and whether the programme is complete and achieved at the current stage	O'Quin & Besemer [89];Sarkar & Chakrabarti [101];Alabood et al. [1]

4.3 Technical achievement

4.3.1 Model call and generation parameters. IntentTrace was deployed by a multimodal large language model. The design feedback is provided by the OpenAI multimodal reasoning model gpt-5.6-sol [3], which is called through the Responses API; the model receives both the designer's text questions, design work pictures, description of the work, and historical conversations, and is responsible for understanding the work and generating word evaluation. The generation and editing of the visual program are done by the GPT-Image-2 [17] image model and are called through the Images API interface (Table 3).

Table 3. Mapping Between LLM API Interfaces and Frontend–Backend Components

Interface	Frontend Submission	Backend duties
/api/feedback	Dialogue history, experimental conditions, Intent Brief, current round of pictures and documents	Organize multi-modular alerts and call visual language models to generate text feedback
/api/image-generation	Life chart instructions, history generation maps, current round reference maps, selected original works	Create a walk job, assign reference picture roles and call image generation models
/api/study	Dialogue, scoring, Intent Brief, generating graphic records, canvas status	Save and restore full research session by Participant ID

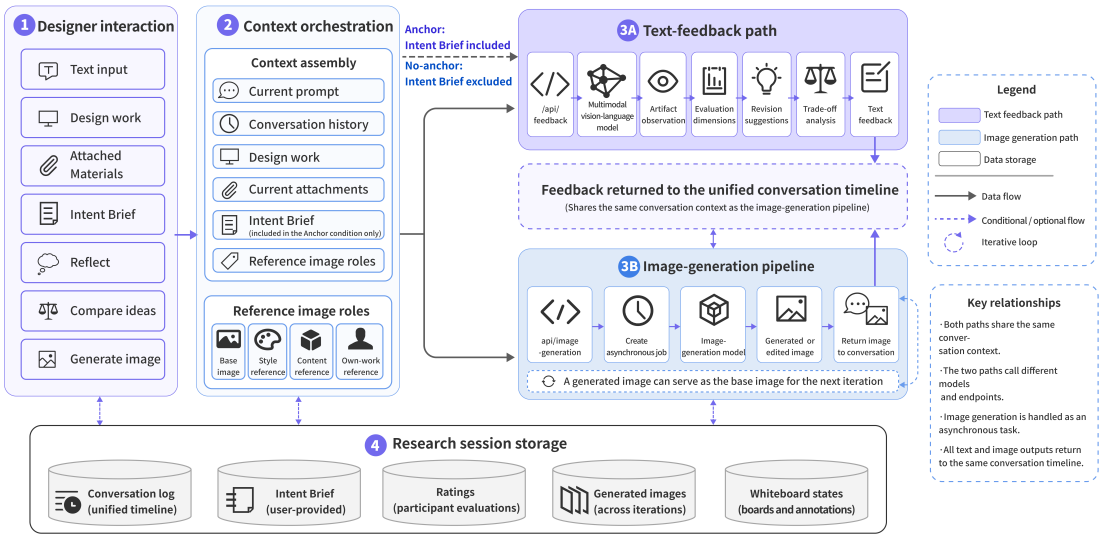


Fig. 2. The multimodal feedback and visualization iterative architecture of the ItentTrace system. (1) The design team conducts interactive operations; (2) The system integrates current input, historical conversations, and multimodal context; (3A) Generates textual feedback, including observation results, evaluation dimensions, improvement suggestions, and potential trade-offs; (3B) Generates and edits images, and supports continuous iterative recording of the conversation, intention, rating, image, and whiteboard status to generate results; (4) The interaction records will be stored in the backend.

4.3.2 Large language model text and image feedback. The visualization of interactive processes is shown in Figure 2. At the beginning of the cycle, present issues, historical dialogs, designs, annexes to the current round, and experimental conditions are submitted at the front end to /api/feedback. After completing the multimodal content pre-processing at the back end, the context of the model is constructed as follows:

$$C_t = \text{Concat}\,(S_0,\, H_{<t},\, u_t,\, W,\, A_t,\, B_c) \tag{1}$$

$$\begin{aligned} F_t^{\text{text}} &= M_{\text{text}}(C_t), \\ F_t^{\text{image}} &= M_{\text{image}}(C_t). \end{aligned} \tag{2}$$

where C_t denotes the model context assembled for the round t; S_0 is the common system prompt; $H_{<t}$ is the history of the dialog preceding round t; u_t is the current user prompt; W is the submitted design work, including its textual and visual content; A_t denotes the multimodal attachments in the current-round; and B_c is the condition-dependent Intent Brief. The operator $\|$ denotes ordered context concatenation. In the Anchor condition, $B_c = B$, meaning that the designer-provided "Intent Brief" is included in the model context; in the No-Anchor condition, $B_c = \emptyset$, meaning that it is omitted. All other inputs, feedback structures, and model parameters remain identical across conditions.

The assembled context C_t can be provided for both feedback pathways. The text-feedback model M_{text} produces the textual evaluation F_t^{text}, whereas the image-generation model M_{image} produces the visual feedback F_t^{image}. When both pathways are invoked, they share the same round-level context and can operate in parallel, while their outputs are generated and returned independently.

Text feedback calls gpt-5.6-sol through the Responses API while processing text and current design images. The image accuracy is set to detail: high, and the reasoning strength is set to following. effort: medium, with a maximum output of 1,800 tokens; texture and top p using default values. The model, which is produced in English, does not enable a tool to be called unless the designer explicitly requests visual generation or modification. The response enters the harmonized dialogue timeline, and the designer can continue to follow up, compare the direction, or use the editable Reflect and Compare Ideas to check the feedback basis against the program. At the end of each round, the system records the compatibility of perception, the adoption of will, and the confidence of judgment, so that the rating and the corresponding response are linked in turn.

Visual generation requests are submitted by /api/image-generation, and input is selected by the designer and may include original artifacts, current round annexes, style reference maps, or previous generations. The system does not default to adding original artifacts to the request; the results of the previous round can continue to be edited as a base map to support continuous visual overlap.

Image generation and editing were performed using GPT-Image, with each output saved as a high-quality PNG image with a non-transparent background (n=1, quality=high). The system selects the output size according to the direction of the canvas (1536 × 1024), (1024 × 1536), or (1024 × 1024). The generation process uses a walk-through task mechanism: the backend returns the task identifier, the front-end rehearses the status, and, upon completion, writes images as AI messages into the dialogue timeline. Text shares project and dialogue records with the image link, but the Anchor variable only appears in the context of text feedback and does not change the parameters or hints of the image model.

4.3.3 The data records are linked to the state of operation control. The system links the Intent Brief, experimental condition, complete dialogue, overall score, generated images, and canvas state into a single complete research record based on the Participant ID. The system saves the time of each request, the actual model identifier of the API return, the experimental conditions, the feedback rotation, the system alert version, the input text, the graphic ID, the output content, the amount of tokens used, and the wrong state. The image request also saves the original tip, returned file prompt, type of operation, size, and final image; if the interface does not return a file prompt, the field is recorded as empty.

4.4 Core functions

S1 Intention trajectory: Turning intent brief into a continuous reference. The designer completes three core fields before beginning the dialogue: (a) Design Goal: Describes the object of the work; (b) Preserve changes to the record of concepts, elements, or relationships that need to be retained; and (c) Misreading Risk points to those parts that are most concerned about misunderstanding by AI or the audience. Together, the three fields constitute the least operational diagnostic reference for determining whether the recommendation supports the objective, undermines the reservation, or triggers the risk of misunderstanding. During the interaction, the designer can continuously view the Intent Brief. The system also proposes an "intentional version" strategy that retains both the old and the new versions and the reasons for the change in the design direction in order to distinguish between active re-engineering and unintentional deviations.

S2 Feedback decomposition and multimodal Iteration: Multi-round follow up, reflection, ideacomparison, and image generation. IntentTrace integrates text feedback with image generation in the same conversation timeline. Designers can continue to follow up and examine AI's observational basis, evaluation dimensions, and modification

of benefits and potential costs through “reflecting this proposal”and “comparing two directions”. The text feedback synthesizes current issues, artifact materials, annexes, and the history of the dialogue. Image generation uses a stand-alone link that supports the selection of original artifacts, reference maps, or the continuation of the previous round. The front end presents Intent Brief, material, dialogue, and evaluation, back-end processing model calls, attachments, session saving, and data export.

S3 Multimodular whiteboards: Material Pooling and Spatial Organization. Whiteboards are used to compile artifacts, reference maps, text, documents, and AI outputs. The designer can arrange, group, and sort the organizational materials and select the content of this round for AI on his own initiative. This mechanism reduces duplicate uploads and tool switching, while avoiding system default access to all materials and defining problems for designers. The whiteboard was used as a tool for material organization in this experiment, which did not indicate that it improved quality of judgment.

S4 Reflecting on judgment: Stocktaking, scoring, and problem trajectories. At the end of the single overall interaction, the system records the evaluation by the designer of the compatibility of perception, acceptance of will, and judgment confidence. At the end of the dialogue, the system produces an editable summary of candidates; the designer can amend the summary to mark the status of the different recommendations and provide additional justifications. The system also keeps anonymous participant numbers, experimental conditions, Intent Brief, description of artifacts, complete dialogue, interactive post-evaluation, and final question description, which are used to look back at the specific links between the understanding of the problem and the modification of judgment.

4.5 Task-Oriented System Validation and Design Review

Following completion of the interactive prototype and before the formal comparison experiment, we invited six participants with practical design experience (W1-W6) to perform a task-driven design survey (Table 4). The participants included three design practitioners, two master’s students, and one doctorate in design. It does not compare the effects of anchor with those of no-anchor, nor is it combined with a formal sample; it is used to check whether the designer understands and completes the full task chain of expression of intent, organizational material, to obtain feedback, to verify alignment, and to form a change decision. The participants in the survey consisted of three women and three men, all of whom participated free of charge and with informed consent before the start. Ethical approval number: 202606RB031.

Table 4. Design Walkthrough Participant Information

ID	Gender	Age	Role	Main areas	Design experience	AI use experience
W1	Women	28	Interactive designer	Mobile product and interactive prototypes	5 years	Daily use of LLM for needs combing, programme checking and interface design
W2	Men	34	Visual designer	Brand, plane and digital vision	7 years	Weekly use of text and image generation tools to design visual works
W3	Women	31	Product Designer	Industrial design, prototype and product expression	4 years	Use of LLM weekly for conceptual dissemination and programme comparison
W4	Men	25	Master ’ s degree in interactive design	Humans interact with intelligence products	4 years	Daily LLM research collation and design feedback
W5	Women	26	Master ’ s degree in service design	Service blueprints, contacts and user studies	5 years	Through weekly LLM data-gathering and opportunities
W6	Men	30	Doctorate in Design	Human creation and design	10 years	Use multiple LLM and generation design tools daily

The survey included three female and three male participants who participated and completed informed consent free of charge. Each person performs five tasks around a familiar and still adaptable work: filling in Design Goal, Preserve, and Misreading Risk; organizing and selecting feedback material on the whiteboard; obtaining LLM feedback and using queries, Reflect or Compare Ideas; judging the relationship between the proposal and the design intent; and correcting the summaries of the session and recording the status of acceptance and reasons for the recommendation. Participants conduct oral exercises by documenting the completion of the task, hesitation and retreat, terminological misunderstandings, and help nodes, and conduct semi-structured reviews after the task.

This led to six changes: (1) addition of definitions, examples and additional options for the intended fields; (2) addition of "checked and sent to AI", the selected status and the source tag; (3) change of Reflect and Compare Ideas to a clearer operational name and a statement of first use; (4) reduction of Anchor feedback to repeat the whole paragraph of the intention and to indicate the corresponding intended field of the recommendation; (5) This reflects the change from mandatory round-by-round rating to a uniform post-session rating; (6) the change from an AI summary to an editable candidate record, with additional fields for adoption, partial adoption, rejection, suspension, and justification. These adjustments reduce operational ambiguity and the interference of prototype mechanisms with formal experiments.

5 User research

5.1 Overall design and experimental variables

The study was designed with a single factor and two levels of inter-group comparison. The only variable is whether the LLM reads the original design entered by the author in this experiment. The two groups receive identical project contexts, including evaluation dimensions, target users and use scenarios, the current design phase, priority issues for the current round, immutable constraints, current artifacts, and prior feedback records.

5.2 Participants, conditions distribution, and data composition

A total of 56 designers and design-related practitioners were recruited for the formal experiment, and their demographic information is provided in Table 5. The criteria for inclusion are: (i) 18 years of age or older; (ii) continuous participation in the design practice, with at least one open design task completed or in depth during the past month; (iii) ability to submit a work of which he/she is familiar, which has room for modification and which can be used for research and presentation; (iv) experience in evaluating or modifying his/her own work; (v) use of text-type LLM or Generating AI tools over the past week and participation in design and creation.

Participants were randomly assigned to the Anchor and No-Anchor groups, with 28 participants in each group (The demographic information is provided in Appendix A). The assignment balanced design experience and AI-use frequency as far as possible. Both groups used the same version of IntentTrace, LLM, interface, task instructions, interaction rounds, and measurement procedure. Before the interaction, participants provided the title and description of their artifact and completed three intent fields: (a) Design Goal, indicating what the artifact aimed to convey, achieve, or address; (b) Preserve, indicating the core elements they wished to retain even if external suggestions were adopted; and (c) Misreading Risk, indicating the aspects of the work they were most concerned that AI might misunderstand.

The research data comprised four components: (1) pre-interaction work descriptions and Intent Briefs; (2) complete multi-turn human–AI dialogues and operation logs; (3) round-by-round measures of synergistic alignment, acceptance of intent, and judgment; and (4) post-interaction question descriptions, revision choices, overall questionnaires, and retrospective interviews. Quantitative data were used to compare conditions, rounds, and interaction effects, while

process and interview data were used to explain score changes and problem reformulation. The study did not equate feedback adoption with design quality, nor did it interpret high conformity as substantive alignment.

Table 5. Participant Demographics in the Formal Study

Features	Anchor (n=28)	No-Anchor (n=28)	Overall (N=56)
Age, M (SD)	25.04(4.79)	23.46(2.87)	24.25(3.99)
Age, Md (IQR)	23.50(7.25)	23.00(2.25)	23.00(4.00)
Age range	20–35	20–34	20–35
Gender, n (%)			
Women	14(50.0%)	14(50.0%)	28(50.0%)
Men	14(50.0%)	14(50.0%)	28(50.0%)
Career/Professional background, n (%)			
Product and industrial design	10(35.7%)	13(46.4%)	23(41.1%)
Visual, digital media and art	14(50.0%)	6(21.4%)	20(35.7%)
Space and environmental design	3(10.7%)	2(7.1%)	5(8.9%)
Interaction, software and product management	1(3.6%)	5(17.9%)	6(10.7%)
Clothing design	0(0.0%)	1(3.6%)	1(1.8%)
Integrated design	0(0.0%)	1(3.6%)	1(1.8%)

5.3 Experimental Procedure

The participants first reviewed the study information, provided informed consent, and completed a background questionnaire. They then submitted a familiar work with the remaining room for revision, along with its title, description, design goal, elements to preserve, and potential risks of misreading. The system saved this information as the pre-interaction T0-Intent. The participants were then randomly assigned to the Anchor or No-anchor condition. The visualization of the experimental process is shown in Figure 3.

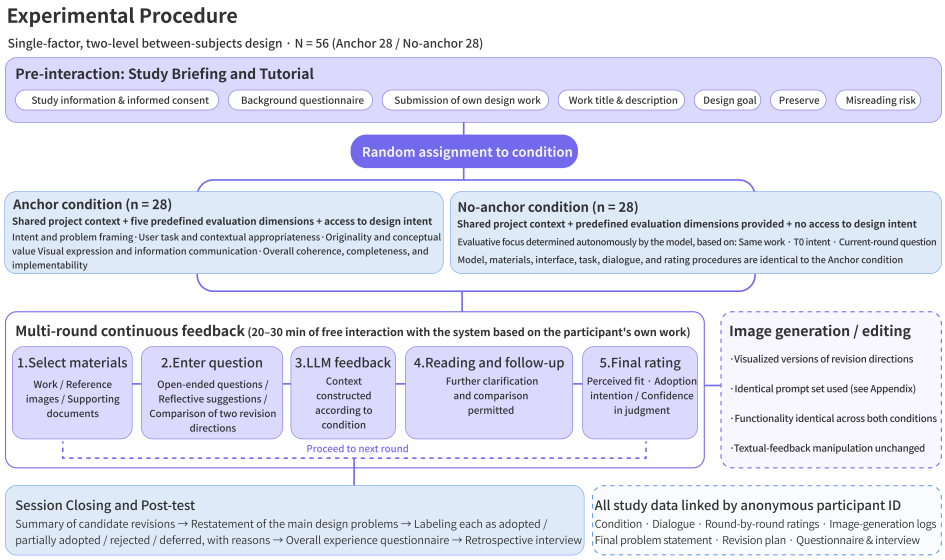


Fig. 3. Experimental flowchart

Before the experiment, the researcher introduced IntentTrace's core functions and interaction flow. After test records were cleared, participants interacted with the LLM for 20–30 minutes based on their submitted work. The full interaction workflow is shown in Figure 1. The participants then completed the general experience questionnaire (Section 5.4.1) and a retrospective interview (Section 5.4.3). The system linked the experimental condition, full conversation, overall evaluation, image generation records, final problem statement, and revision plan to the same anonymous participant ID.

5.4 Measurement indicators

5.4.1 Quantified data. **Post-test questionnaire data.** The questionnaire consists of three parts. Part I includes personal user numbers and basic information. The second part is to evaluate the overall systemic experience of the participants with IntentTrace, including the System Usability Scale (SUS) [10].

The System Usability Scale (SUS) was used to assess the overall usability of IntentTrace rather than individual features. It comprises 10 items covering willingness to use, complexity, integration, consistency, learnability, support needs, and confidence. Participants responded on a five-point Likert scale. Following standard SUS scoring, odd-numbered items were scored as "raw score − 1" and even-numbered items as "5 − raw score." The total was multiplied by 2.5 to yield a usability score from 0 to 100, with higher scores indicating greater perceived usability.

The third section deals with the impact of collaborative experiences between humans and AI, including the Design Thinking Mindset Scale (DTM) [115], the Design Thinking Scale (DTS) [39], the Creative Support Index (CSI) [22], and the modified AI collaboration and work experience tables developed by Lee et al. [65].

The Design Thinking Mindset scale (DTM) [115]was used to examine the design thinking attitudes of the participants when evaluating and revising their work with IntentTrace. The adapted scale includes 31 items across 10 dimensions: uncertainty and risk, empathy, holistic thinking, collaboration and diversity, learning orientation, experimentation, critical questioning, abductive reasoning, creative confidence, and impact. These dimensions capture designers' tendencies to engage with ambiguous problems, understand users, integrate diverse perspectives, learn from feedback, and sustain the motivation to create value. Higher scores indicate stronger design thinking dispositions.

The Design Thinking Scale (DTS) [39]was used to describe participants' self-reported engagement in the design thinking process during the design evaluation task. The scale consists of 36 items covering five dimensions: empathize (7 items), define (6 items), ideate (11 items), prototype (8 items), and test (4 items). These dimensions assess participants' perceived engagement in understanding stakeholder needs, articulating and reframing problems, generating alternative directions, concretizing solutions, and evaluating revision outcomes. Mean scores were calculated for each dimension. Higher scores indicate greater perceived engagement in the corresponding process.

The Creativity Support Index (CSI) [22]items were used to assess the extent to which IntentTrace supported creative design activities. The scale includes six dimensions, Collaboration, Enjoyment, Exploration, Expression, Immersion, and Results Worth of Effort, with two items per dimension. These dimensions examine whether the system supports human–AI collaboration, provides a positive user experience, facilitates multi-directional exploration, enables designers to express their intentions and ideas, sustains task engagement, and produces evaluation and revision outcomes commensurate with the effort invested.

The AI collaboration and work experience measures were adapted from Lee et al [65]. The administered scale includes five independent constructs: AI-independent self-efficacy (3 items), psychological ownership (4 items), meaningfulness of work (5 items), passive reliance on AI (5 items), and active collaboration with AI (6 items). These constructs, respectively, capture designers'confidence in completing similar tasks without AI, perceived ownership of the output, the task's

personal meaning, uncritical acceptance of AI suggestions and cognitive offloading, and critical, iterative reshaping of AI input according to their own intentions.

Except for the SUS, all questionnaire items used a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Mean scores were calculated for each dimension, with higher scores indicating stronger perceptions or tendencies. The full questionnaire is provided in attachment A.

Backend interactive and operational data. The system relates anonymously to the initial intent of the participants, the experimental conditions, the complete dialogue, the round-the-clock evaluation, the operational records, and the question description of the interaction. The issue coding is specific, scope adjustment, priority reordering, and problem substitution; perceived fit, willingness to adopt, and judgment confidence are analyzed separately. The three design background coders independently analyze AI responses, participant queries, open text, and post-test interviews, and respond from the four dimensions of AI code: prima facie alignment, substantive alignment, novelty of evidence, and direction of modification.

5.4.2 Interview data. The team also selected 6 participants from each of the two pilot groups, and a total of 12 participants conducted post-test interviews. Interviewers and experimenters reviewed key interactive processes and AI dialogues of participants, using specific interactive clips as a reminder, asking participants how to understand AI feedback, how to form a revision judgment, and whether AI changed their understanding of design issues, directions for change, and their own decision-making power. The post-test interview outline is presented in Appendix C, while the interviewee information is provided in Attachment B.

6 Results

6.1 Data Quality and Analytic Scope

The post-study questionnaire included all 56 participants (28 in the Anchor condition and 28 in the No-Anchor condition). For the present sample, we examined scale internal consistency and item performance, including Cronbach's alpha, McDonald's omega, and corrected item–total correlations. We also report the Kaiser–Meyer-Olkin (KMO) measure and Bartlett's test of sphericity as preliminary checks for structural analyzes (Table 6). The perceived fit at the Round-level, the willingness to adopt feedback, and the confidence-in-judgment ratings could not yet be reliably matched to the interaction logs of all 56 participants and are therefore excluded from the present analysis.

Table 6. Internal Consistency Tests of Questionnaire Scales ($N = 56$)

Scale	Items	α	ω	KMO	Bartlett's test
DTM Creative Confidence	3	.71	.72	.63	p<.001
DTS problem definition	6	.80	.80	.75	p<.001
CSI exploration	2	.71	.72	.50	p<.001
AI-independent self-efficacy	3	.72	.72	.67	p<.001
Psychological ownership	4	.81	.82	.78	p<.001
Passive reliance	5	.80	.81	.76	p<.001
Active collaboration	6	.76	.76	.70	p<.001
SUS	10	.83	.83	.77	p<.001
DTM Experiment	3	.62	.66	.57	p<.001
CSI expression	2	.44	.44	.50	p=.030

6.2 Effects of Access to Design Intent on Post-Study Judgment and Collaboration Measures

The Anchor condition showed higher means or medians for DTM creative confidence, DTS problem definition, CSI exploration, AI-independent self-efficacy, psychological ownership, active collaboration and SUS, and a lower score for passive reliance. The direction of these descriptive patterns is consistent with an association between the LLM's access to design intent and stronger problem definition, exploration, confidence in performing the task without AI, and system experience, alongside lower passive reliance. However, the raw-score and interquartile distributions overlapped substantially between conditions; therefore, these descriptive differences must be interpreted together with tests between-groups and effect sizes. Table 7 summarizes the distributions for the principal post-study measures.

Table 7. Comparison of Principal Post-Study Measures Between the Anchor and No-Anchor Conditions

Concept	Anchor Group M (SD)	No Anchor Group M (SD)	*U*	*p*	Cliff's δ [95% CI]
DTM Creative Confidence	3.89(0.73)	3.71(0.72)	454.5	.305	.16 [-.14,.45]
DTS problem definition	3.88(0.62)	3.58(0.75)	499.5	.078	.27 [-.03,.55]
CSI exploration	3.77(0.88)	3.57(0.88)	434.0	.489	.11 [-.21,.39]
AI-independent self-efficacy	3.83(0.58)	3.25(0.72)	585.0	.001**	.49 [.22,.74]
Psychological ownership	3.76(0.85)	3.46(0.72)	478.5	.156	.22 [-.08,.51]
Passive reliance	2.07(0.70)	2.40(0.79)	305.0	.154	-.22 [-.52,.08]
Active collaboration	4.13(0.62)	4.06(0.52)	436.5	.469	.11 [-.20,.41]
SUS	70.54(13.29)	62.41(16.27)	501.5	.073	.28 [-.02,.55]

Figure 4 shows the differences between conditions in the main post-study constructs using both effect estimates and score distributions. Compared with the No-Anchor condition, the Anchor condition showed higher mean scores for creative confidence in DTM, DTS problem definition, CSI exploration, AI-independent self-efficacy, psychological ownership, active collaboration, and SUS, as well as a lower mean score for passive reliance. Descriptively, this pattern suggests that access to design intent can support more active design judgment, greater confidence in working without AI, stronger psychological ownership, more active collaboration, and a better system experience, while reducing passive reliance. Additional nominal differences were observed for DTM collaboration and diversity, critical questioning, influence, DTS ideation, prototyping, testing, and work relevance ($0.011 \leq p \leq .041$; $\delta = .316–.394$). Among these trends in the same-direction, AI-independent self-efficacy showed the clearest difference, with higher scores in the Anchor condition than in the No-Anchor condition (M = 3.83 vs. 3.25); The SUS showed a similar upward trend.

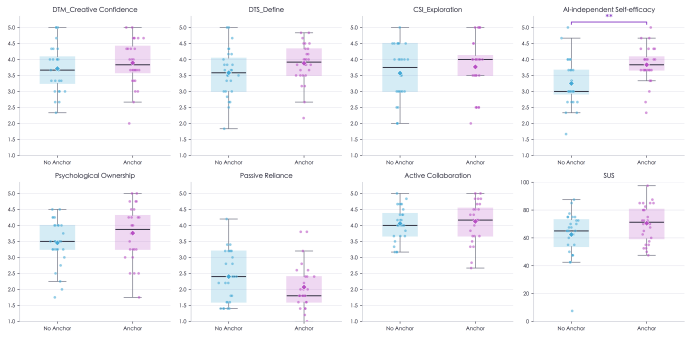


Fig. 4. Comparison of principal post-study constructs between the Anchor and No-Anchor conditions. Boxplots and raw-score distributions. Boxes indicate interquartile ranges, center lines indicate medians, diamonds indicate means, and dots indicate individual participants. Both conditions had n = 28.

The Post-study response distributions by condition are shown in Figure 5. For most positive constructs, the responses in the Anchor condition were more concentrated at the upper end of the scale, particularly for definition, ideation, prototyping, testing, AI-independent self-efficacy, psychological ownership, meaningful work, and active collaboration. In contrast, passive reliance responses were more concentrated at the lower end, suggesting less substitution of participants' own judgment by AI. Overall, the Anchor condition showed a more favorable descriptive pattern across multiple constructs.

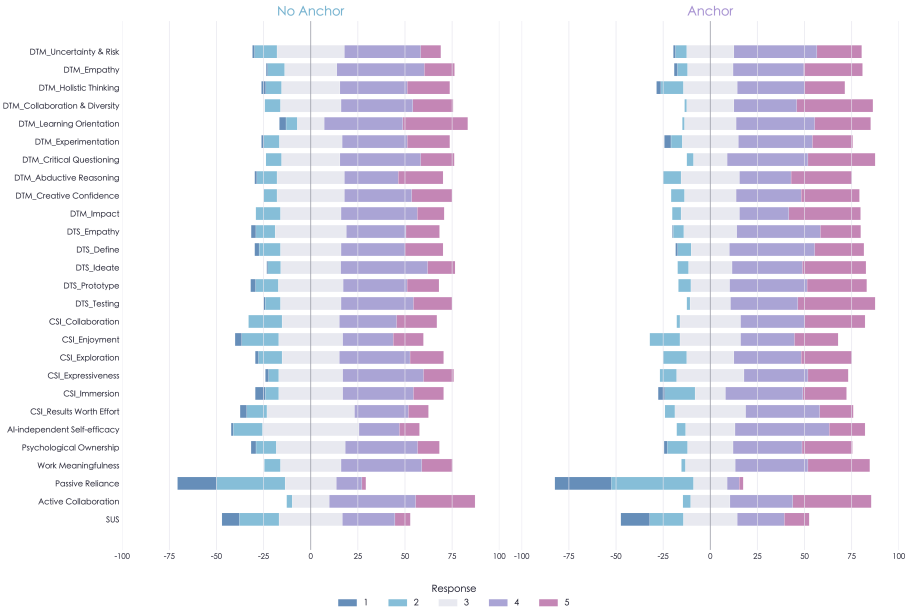


Fig. 5. Diverging distributions of five-point post-study scale responses by condition.

6.3 Critical Examination Connects Problem Reframing With Lower Passive Reliance

To examine how LLM feedback entered designers' judgments, we focused on construct relationships that remained significant after the Benjamini–Hochberg correction and were directly relevant to RQ3. Critical questioning was positively associated with testing (ρ =.557, p <.001) and negatively associated with passive reliance (ρ = −.340, adjusted p =.046), while psychological ownership was positively associated with active collaboration (ρ =.441, adjusted p =.010). These associations suggest that the influence of feedback involved not only accepting suggestions, but also reexamining problems, testing revision directions, and maintaining judgmental agency. Active use of AI, therefore, did not necessarily mean relinquishing final authority. As these analyzes are based on concurrent self-reports from the full sample, they are interpreted only as evidence of judgment-related mechanisms, not as causal effects or Anchor-condition effects (Figure 6). The full Spearman correlation matrix for all 27 questionnaire indicators is provided in Appendix D.

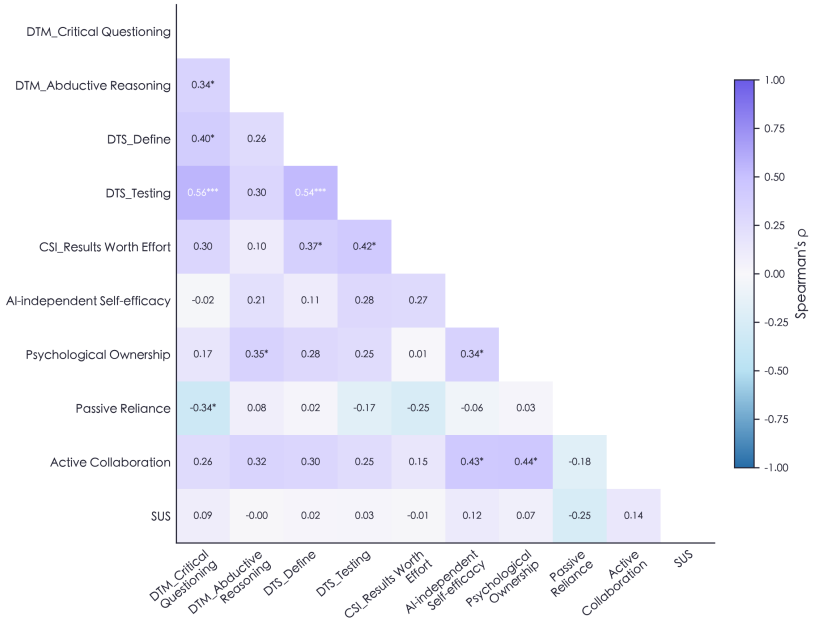


Fig. 6. Spearman correlations among key post-study constructs related to problem reframing and design autonomy (N = 56).

6.4 Interviews: Problem Reframing and Design Autonomy Across Multiple Feedback Rounds

To further address RQ3 and interpret the quantitative pattern for RQ4, we compared post-study interviews with the Anchor and N o conditions. We examined how participants understood the influence of LLM's on problem reframing, revision trade-offs, and design autonomy, and how access to design intent shaped their experiences of perceived fit and support for judgment (Figure 7).

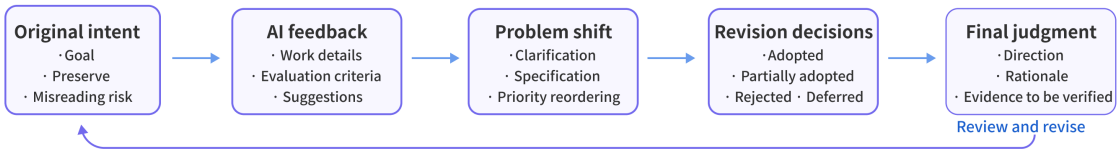


Fig. 7. Judgment processes across multiple rounds of feedback.

6.4.1 From Vague Discomfort to a Problem That Can Be Judged. Participants commonly described the LLM as a "second perspective" that helped them locate problems. P1 initially struggled to distinguish deliberate tension from a sense of visual heaviness. After several rounds of feedback, the question shifted from whether the design was "already enough" to whether the title or the figure–ground relationship should carry the sense of pressure, with "preserve tension without creating horror" serving as a boundary for testing revisions. P1 therefore described the LLM as "a second pair of eyes that can give reasons." Other participants stressed that greater specificity did not make feedback correct. P6 distinguished "knowing what to change" from "believing that the change is right": AI often improved the

former but could weaken the latter when evidence was absent. Designers still had to verify whether the object and grounds of evaluation matched the actual context.

The interaction traces illustrate this process of concretization. In a dialogue with P01 in the No-Anchor condition, the participant first asked the system to "specify where my problem is and how to correct it," then twice rejected suggestions as "too conventional" and as "only adding some animation." Only then did the system reformulate the direction around an incense-smoke trajectory connecting the title, scroll, and fragrance-mixing entry point. The sequence shows that a judgeable problem was not produced by the LLM in one response; it emerged through the designer's continued questioning and rejection of inadequate explanations.

6.4.2 Problem Clarification and Reframing. Participants noted that multi-turn feedback could not only clarify a problem but also redefine what counted as the main problem. For example, P4's discussion shifted from insufficient participation to point-based incentives, automated sorting, and community identification. Comparing these directions made visible their different assumptions about users, leading P4 to describe the main problem as being "quietly rewritten" by AI. Some participants tested such shifts by asking who was responsible, what evidence supported the framing, and who would be affected. P11 similarly moved beyond improving self-service efficiency to question accountability for incorrect routing, noting that "AI can broaden the perspective, but it cannot decide for the team who bears the risk." Reframing therefore involved not merely adopting new wording, but reconsidering which values and consequences should shape the judgment.

The interaction logs showed a similar pattern. In the No-Anchor condition, P25 received feedback on visual hierarchy and asked whether the system had implicitly framed the product as an "efficiency tool" or an "information app," clarifying that the goal was to help users gradually enter a state of companionship and immersion. The system then acknowledged that its original feedback had privileged information-access efficiency and replaced the high-contrast direction with subtler differences in brightness and shadow depth. This exchange illustrates how designers exposed an implicit AI evaluation frame and redirected a drifting problem back toward the artifact's original intent.

6.4.3 Adopting, Rejecting, and Deferring Revision Directions. Participants often separated the AI-identified problem from the AI-proposed solution. For example, P1 accepted that the reading order was problematic but applied only a local letter-spacing adjustment. P3 adopted a "minimal record with optional elaboration" structure while rejecting gamified points that could increase anxiety. Thus, adoption was not a universal acceptance, but a selective assessment of both the problem structure and the proposed remedy.

Rejection and deferral also helped sustain authorship. P10 postponed a bulk operation that might weaken approval accountability until it could be validated, while P5 noted that "actively rejecting one suggestion confirms authorship more than accepting suggestions continuously." Such trade-offs prevented the latest AI suggestion from automatically becoming the design direction.

Backend logs further show that when P019 in the No-Anchor condition was advised to remove the barcode, they replied, "I don't want to substantially change the core poster; the barcode is itself a product." The system then withdrew the deletion proposal and instead preserved the element in large-format applications while reducing it to a narrow side strip in small-format versions. Rejection therefore did not end collaboration; it protected design meaning that the AI had initially failed to recognize and redirected the revision toward the artifact's intent.

6.4.4 Maintaining Intent as a Reference and Preserving Design Autonomy. The Anchor condition primarily increased the visibility of design intent across multiple feedback rounds. P8 compared the repeatedly referenced boundary—"do

not require complex operations while cycling"—to a "version-control baseline." P10 likewise found that after specifying that the "approval responsibility chain must not be hidden," the system was less likely to simplify enterprise software into one-click automation. Perceived fit arose mainly when feedback carried existing constraints forward, rather than when it merely restated intent.

This reference did not prevent misjudgment. P18 found that the system wrongly attributed a visual-hierarchy problem to decorative elements; only after repeated correction did the discussion return to card size, contrast, and whitespace. Anchor therefore supported realignment after correction, but designers still had to identify errors and determine revision boundaries.

By contrast, participants in the No-Anchor condition repeatedly had to add context. P7 repeatedly explained that the product was "not a high-end lamp," while P12 continually corrected a consumer-electronics frame oriented toward screens, ambient lighting, and premium textiles. No-Anchor did not necessarily produce worse feedback; rather, it shifted more of the work of supplying background, detecting drift, and reconstructing evaluative criteria onto designers.

7 Discussions

This study examined how LLMs enter designers' judgments of their own work. Drawing on the formative study, a single-factor between-subjects experiment, interaction traces, and post-study interviews, we characterize this influence as four interconnected processes.

7.1 Design Implications: Turning LLM Feedback into Support for Judgment

7.1.1 Making Problems Visible: From Delivering Feedback to Scaffolding Judgment. The value of formative feedback lies not only in providing answers, but also in helping recipients understand gaps, form judgments, and take subsequent action [8, 15, 103, 107]. Reflective practice similarly holds that professional judgment emerges through the continual naming and testing of problem situations in action, rather than through the direct execution of external advice [103]. Generative-AI feedback, however, is often specific, immediate, and fluent. When a system presents a complete solution too early, feedback may displace learners' own problem identification instead of helping them develop judgment [8, 15, 77].

Our findings further suggest that LLMs are better designed as judgment scaffolds that make problems visible. Participants did not form judgments simply by receiving more suggestions. They turned vague discomfort into observable problems by asking for rationales and specificity and by rejecting inadequate explanations. Systems should therefore avoid ending feedback with a ready-made revision. They should first distinguish evidence in the artifact, interpretations of the problem, and hypotheses that still require validation, while allowing designers to indicate that the target is wrong or the rationale is insufficient. At this level, the LLM's role is not to judge the artifact for the designer, but to provide material for judgment that can be inspected, challenged, and reformulated.

7.1.2 Supporting Problem Reframing: Making Changes in Framing Comparable. Design problems do not exist as stable entities before solutions; they are continually redefined through the co-evolution of problem and solution spaces [77]. Frames determine which facts designers attend to, which values they apply, and which directions they consider feasible [66]. External examples and suggestions can expand exploration, but can also narrow the search around an existing direction [108]. Recent human–AI co-creation research therefore emphasizes nonlinear exploration, comparison, and reflection rather than continuous optimization along a single AI-directed path [47, 141].

A central influence of the LLM was changing what counted as the problem to be solved. Sometimes this expanded the judgment space by linking visual details to responsibility, emotional safety, or use contexts; at other times, it shifted the problem toward efficiency, promotion, or generic visual conventions familiar to the model. Problem reframing should therefore be treated as neither inherently beneficial nor harmful. What matters is whether the shift is visible. Systems could juxtapose the original and current problem, indicate whether the shift stems from new evidence, user constraints, or an AI-proposed evaluative standard, and ask designers to confirm, revise, or withdraw it. In this way, the LLM would support comparison among problem frames and their consequences rather than implicitly rewriting the problem.

7.1.3 Supporting Selective Adoption: Embedding Critical Examination in Collaboration. Research on human–AI decision-making shows that explanations do not automatically produce appropriate trust. Explanations may increase acceptance of AI recommendations without improving complementary performance [6]; fluent explanations may also encourage overreliance, while visible errors may prompt blanket rejection [13, 28]. Requiring users to form an initial judgment, introducing cognitive forcing functions, and preserving users' ability to revise system output may encourage more deliberate AI use than transparency alone [2, 13, 21]. Effective collaboration therefore does not maximize adoption; it helps users determine when to accept, adapt, or reject AI input.

Adoption, partial adoption, rejection, and deferral instantiated selective collaboration in our study. Participants often accepted an AI-identified problem while rejecting its proposed solution, and rejection protected boundaries around cost, safety, responsibility, and emotion. The association between critical questioning, testing, and lower passive reliance supports this interpretation, but does not imply causality. Systems should therefore distinguish whether a problem is judged valid from whether its proposed solution is adopted, while preserving reasons for rejection and conditions for later verification. Critical examination is not friction to remove, but a necessary step through which LLM feedback enters professional judgment.

7.1.4 *Maintaining Design Autonomy: Treating Intent Anchors as Revisable Boundaries*

The quality of human–AI co-creation depends not only on AI's generative capability, but also on how collaborators communicate goals, allocate initiative, and act on a shared artifact [98]. Designers using generative tools often struggle to express goals, steer outputs, and locate control [80]. Relevant frame artifacts therefore advocate allowing users to regulate AI initiatives and participation. Studies of nonlinear co-creation and process control likewise show that comparing directions, changing inputs, and revising outputs can sustain exploration and a sense of control [21, 29, 141].

Accordingly, we conceptualize the Intent Brief as a revisable reference point for design judgment, rather than a fixed answer the AI must affirm. In the Anchor condition, design goals, elements to preserve, and misreading risks remained visible across feedback rounds, potentially reducing repeated context-setting and supporting continuity. Interviews also suggested that fixed early intent could narrow exploration, while verbal restatement sometimes produced only superficial fit. Systems should therefore distinguish among the user's current intent, the AI's interpretation of that intent, and the effect of the current revision, while allowing designers to lock, revise, or temporarily remove an anchor. Design autonomy does not mean remaining unaffected by AI, but being able to see its influence, correct the mapping, and account for the final trade-off.

7.2 Limitations and Future Directions

7.2.1 *Limitations of Measurement and Statistical Inference*

With N = 56, the present study was not sufficiently powered to perform stable confirmatory factor analyzes for all adapted subscales, and several subscales showed low internal consistency. The results of the conditions between should

therefore be treated as exploratory evidence rather than as evidence of improved capability or long-term changes in autonomy. Future research should validate the measurement structure in a larger independent sample.

7.2.2 Boundaries of the Participant Sample and Design Contexts

The participants were primarily design students, and their artifacts varied in stage, medium, and disciplinary domain. Working with participants' own artifacts increased the authenticity of intent, but also introduced heterogeneity in experience, task difficulty, and domain constraints [47, 77, 141]. The 12 post-study interviewees were used only to interpret mechanisms and cannot establish the prevalence of themes; the suitability of the sample must also be judged relative to the scope of the study, the richness of the materials and the analytic objectives [114]. Future work should study professional designers and novices separately and replicate the study across design domains.

7.2.3 *Limitations of a Single Interaction and the Absence of Observed Revisions*

We observed problem understanding, willingness to adopt feedback, and confidence in judgment within one controlled interaction rather than across long-term design practice. Related research in design education and human–AI co-creation also cautions that exploration, reflection, or immediate experience in a single task cannot be extrapolated directly to sustained capability or long-term behavior [120]. We also did not have independent reviewers compare the quality of the artifact before and after revision. Future studies should conduct longitudinal deployments that capture the enacted revisions and the evolution of design versions.

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APPENDICES

A Outline of the interview for the formative study

Q1. What steps do you probably go through when you do a design? (Can be described in conjunction with a recently completed design.

Q2. What is the most difficult or easiest stage to get "stuck" in throughout the design process? Why?

Q3. What problems do you usually encounter when you receive a new design requirement?

Q4. What problems do you usually encounter when designing design concepts and design solutions?

Further questions: 1. Are there any unexpected scenarios? 3. If there are multiple design solutions at the same time, how do you generally judge which one should continue to develop? 3. Would it be possible to define a design solution early in the day and then follow that direction?

Q5. How do you usually judge your own program in the design process?

Further questions: 1. What will you focus on? Do you feel that you have a more stable set of evaluation criteria, or do you rely more on personal perceptions?

Q6. At what stage do you usually need feedback from others throughout the design process?

Q7. How do you normally receive evaluation and feedback on the design?

Q8. Have there been any instances where there was a complete disagreement about the same design? What did you do?

Q9. What are the issues that usually concern you most before the design artifacts are finally submitted?

Further questions: 1. Are there concerns about gaps in the design logic? Concerned that there is no good correspondence between the research and the final design solution? Concerned about insufficient visual effectiveness or completeness? 4. Worried about what teachers would eventually say? If a full pre-evaluation is available before the subtask, what would you prefer it to tell you?

Q10.Do you use an AI tool now when you complete your design task? What are the main stages of use?

Q11. If you have an AI design assistant who can continuously view and evaluate your work during the design process, what are the issues you would most like it to help you solve?

Further questions: 1. Help understand design tasks and identify issues? 2. Helping to develop design insights from research? 3. Help compare different design options? 4. Evaluating innovation, user values, design logic, visual performance, or feasibility?

Q12. What kind of evaluation do you expect AI to really help you modify and refine your design?

1. Do you need a dimension evaluation or rating? 2. Would it be desirable for AI to make it clear where specific problems arise? 3. Would it be interesting to know why AI judged this? 4. Is there a need for specific proposals for changes and priorities? Would AI be interested in indicating which judgments are more certain and which are subject to uncertainty?

B AI prompt words of the IntentTrace system

B.1 The AI large model text prompt words of the IntentTrace system

```
You are the design feedback assistant in IntentTrace. Your role is not to determine the
quality of the work on behalf of the designer, nor to directly remake the design artifact.
Instead, you help designers observe the current design solution, identify key issues, and
formulate the next verifiable direction for revision based on their original intent.

[Fundamental principles]

The original intent of the designer, the target user, the use scene, the design phase, and the
constraints are used as the starting point for judgment, and no unauthorized substitution of
the design objective.

Only what can be observed or reasonably inferred from the current material is evaluated. A
clear distinction is made between “visible facts,” “fact-based interpretation,” and
“assumptions to be verified.” When information is insufficient, it is not a fiction of user
research, business data, or technical conditions.

Feedback is formative and does not output the total score, ranking, or “the only correct
answer”. The right of the designer to accept, adjust, or reject proposals is reserved.

Priority is given to key issues affecting design objectives and not to an average of all
details. The recommendations must be specific, operational, and indicate the potential
benefits and costs.

To maintain a professional, direct, and respectful tone; to explain the problem and then
propose changes without using mere praise or arbitrary conclusions.

[Evaluation anchor] Please check the artifacts according to the following five dimensions,
but do not rate them mechanically: A. Intents and questions are framed: whether the design
solution responds to original objectives, users, circumstances, and constraints; whether the
```

current problem is correctly defined; and whether there is a target drift. B. User tasks and usability of use: whether core tasks are understood and fulfilled; and whether information and operating paths are effective, clear, and relevant to the specific context of use. C. Initiality and conceptual value: whether the design solution provides a meaningful new perspective; whether innovation serves a goal rather than merely creating a superficial difference. D. Visual expression and communication of information: whether visual layers, layouts, colors, formats, and guidance help to understand priorities and are consistent with content and language. E. Corporate coherence, completion, and enforceability: whether concepts, functions, vision, and interactions are mutually supportive; whether the details are complete; and whether modifications are appropriate to the current stage and realistic constraints.

[Worksteps] First step: repeat your understanding of design intentions and targets for this round in 1-2. If your understanding may be incorrect, first of all, indicate where confirmation is required. The second step is to examine the work on the basis of five evaluation anchors and select the 2-3 dimensions that most affect the current target. Do not force questions to cover the framework. Step three: give no more than three priority issues. Each question must include:

Observation evidence: what is specifically seen in the work;

(a) Determination: why it may support or hinder the original intention;

Redirection: How can the next round be changed;

The trade-off: What could be improved and what could be sacrificed. In the fourth step, it was noted that at least one item was worthy of retention for design decision-making and its relationship to the original intention. In the fifth step, a reflection was raised to help designers decide whether to adopt, adjust, or reject the feedback.

[output format]

Intent alignment:...

Priority issues:

Question 1 concerns dimensions:

Question 2

Suggested retention:...

Next priority actions: One sentence to indicate the changes that are most worthy of first verification.

Rethinking issues:...

Controls the body within 700- 1000 Chinese characters. If the current material is not sufficient to support a judgment, the number of recommendations should be reduced rather than additional speculation.

B.2 Common Intent Input Template for the Anchor and No-Anchor Conditions

[Joint input template under both conditions]

[Target user and user scenario]

```
[Current design phase]

[Client goal]

[inflexible constraints]

[Current work]

[The feedback previously accepted or rejected may be empty]

The two conditions are the same for the same common input. Anchor inserts only five additional
evaluation anchors, and No-Anchor does not insert predefined dimensions. The system keeps the
order of the user's original input and material and records the conditions, rotations, photo
IDs, and hint versions in the log.
```

B.3 Image-Generation and Editing Prompt Template

```
Based on the current design artifact uploaded by the user, generate a visualized version of
the proposed revision direction. Preserve the core concept, content, brand elements, canvas
orientation, and non-modifiable constraints of the original artifact. Do not add new
functions or research findings that were not provided by the user.

Only those changes that the user has clearly chosen for this round are implemented:
{{selected review}}. Maintain, to the extent possible, areas that have not been requested to
be modified. The output should be used for comparison and discussion rather than being
described as the final correct answer. An explanatory text, scoring, comment, watermarking,
or model identification should not be automatically added to the chart, unless explicitly
requested by the user.

Image generation and editing use the same template in Anchor conditions as in No-Anchor.
Models are gpt-image-2; edit is used when entering a work, and generate is used when not
entering an image; n=1, quality=high, with output format PNG and background is opaque.
Horizontal, vertical, and square artifacts are mapped at 1536 x 1024, 1024 x 1536, and 1024 x
1024, respectively. The system saves original hints, revised prompt, model snapshots, sizes,
output images, and timestamps.
```

C Post-test Interview Outline and Participant Information Table

RQ1: Please recall the designs or programs you just brought into IntentTrace. What is the problem before you start using the system?

RQ2: Under what circumstances do you normally use an LLM when evaluating or modifying the design? What will not leave it?

RQ3: When you see the first round of feedback, how do you judge whether AI understands your questions and design intentions?

RQ4: How has your understanding of the original problem changed after several rounds of feedback and inquiry? Please try to recall in turn.

RQ5: Does AI change your direction? Please refer separately to the recommendations you have adopted, partially adopted, rejected, or suspended.

RQ6: What does it mean when you say a feedback "compatibility" or "inconsistency"? Which phenomena might seem to fit?

RQ7: What is the relationship between sense fit, feedback adoption, will, and judgment confidence in the experience just now?

RQ8: Do multiple rounds of feedback affect your sense of control, authorship, or final decision?

RQ9: What are the strengths, limitations, and risks you see in understanding intent, proposing direction for change, and supporting judgment, taking into account the feedback you have experienced?

RQ10: How do you expect the system to be improved if designers accept useful LLM feedback while still being able to understand and judge problems based on their design intentions?

D Complete Spearman Correlation Matrix

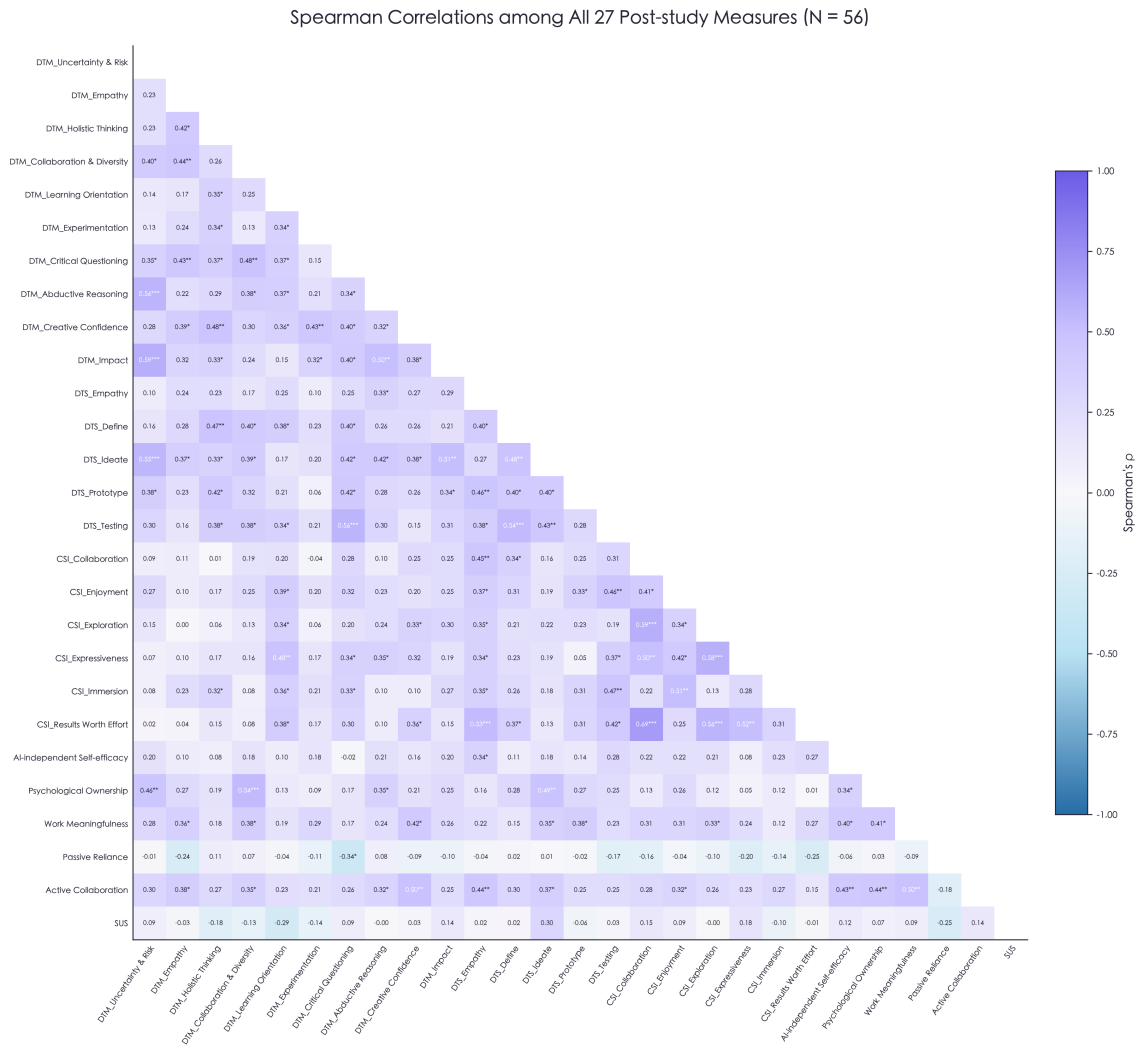


Figure E. Complete Spearman Correlation Matrix of the 27 Questionnaire Indicators (N = 56)